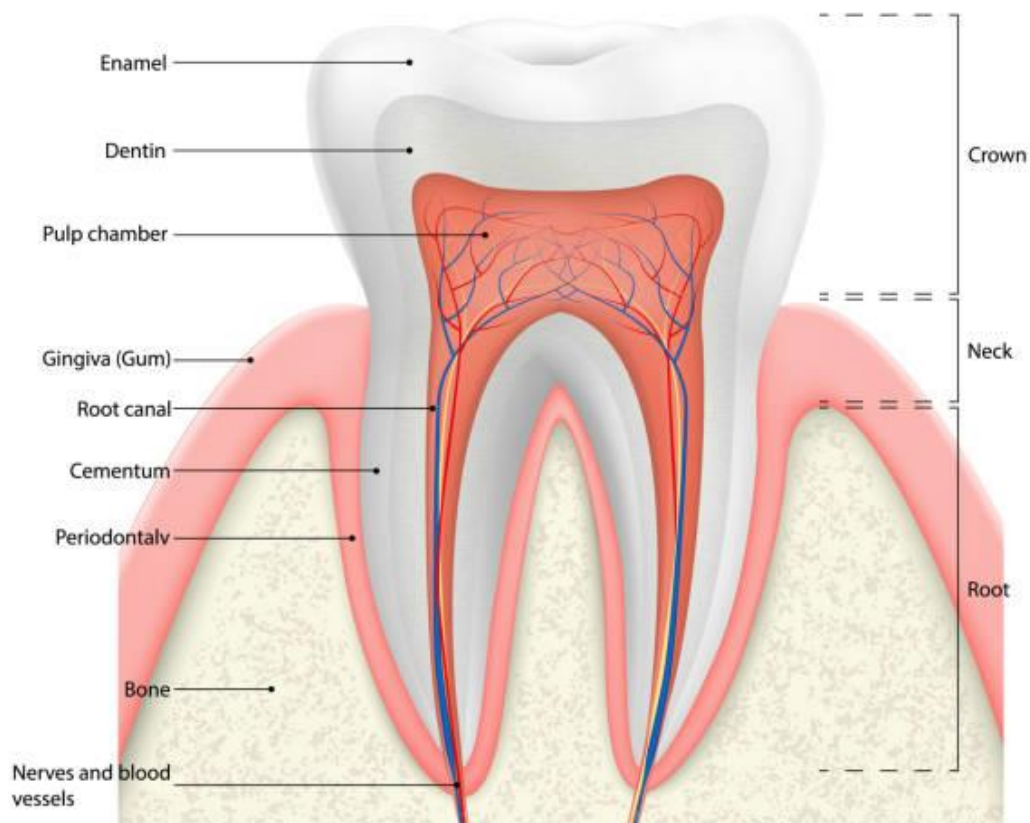




Diseases of the dental pulp

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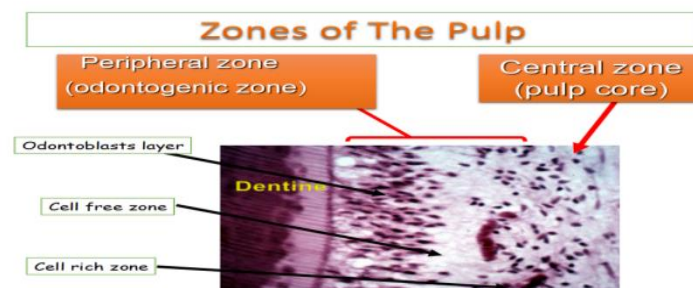
**Third Year
2026- 2027**

Disorders of the dental pulp

❖ Dental pulp

Dental pulp can be defined as a richly vascularized, innervated and unmineralized connective tissue enclosed by dentin with communication to the periodontal ligament by apical foramen and accessory canals. The embryonic origin of dental pulp from dental papilla which originates from neural crest cells.

During histological examination, four unique zones can be distinguished: Odontoblastic zones: it is representing the peripheral aspect of pulp, which contains the odontoblast cells body. Cell-free zone or zone of Weil: This zone contains many bundles of fibers, numerous capillaries and nerves but does not contain any cells. Cell-rich zone: it contains several fibroblasts which represent the predominant cells type of pulp and also contains undifferentiated mesenchymal cells which can be differentiated into different pulp cells. Pulp core: This zone mainly contains the main blood vessels in the pulp and Raschkow plexus of nerves .



Dental pulp like other connective tissues throughout the body; it reacts to noxious stimuli by an inflammatory response. This response is not significantly different from that seen in other tissues, the final result can be different because of certain anatomical features of the pulp which includes: -

1-The pulp is enclosed within the calcified walls of the dentin, which precludes the excessive swelling of the tissue that occurs in hyperemic and edematous phases of inflammation in other tissues.

2-Absence of collateral circulation.

3-Lack of proprioceptors making pain localization to correct tooth is a diagnostic challenge.

Causes of pulpal disease:

I-Living irritants (Bacterial)

Bacteria can gain access to the pulp through dental caries, fracture of the tooth exposing the pulp, or haematogenous infection (Anachoresis).

II- Non living irritants:

1-Physical

a-Thermal

- Heat from cavity preparation
- Exothermic heat generated during cement setting

b. Electrical (galvanic current from dissimilar metallic fillings)

2-Mechanical:

a-Traumatic accident

b-Attrition and abrasion

c- Cracked tooth syndrome; particularly in posterior teeth, which may develop cracks or split under masticatory stresses.

3.Chemical:

a- Erosion

b- Inappropriate use of acidic dental material, for example, sterilizing agents, monomers present in composite restoration, or arsenic present in silicate cement.

4. Idiopathic: Aging, internal and external resorption.

5. Barodontalgia: It's defined as tooth pain resulting from change in barometric pressure. "Baro" means pressure, and "odontalgia" means tooth pain. Is a condition in which there is toothache in persons flying at high altitudes and underwater divers. It is usually associated with recently filled teeth or teeth affected by hyperemia.



N.B: Dental *attrition, abrasion, and erosion* are types of dental wear. However,

abrasion (caused by contact between teeth and other materials), **attrition** (tooth-to-tooth contact), and **erosion** (dissolution of hard tissues by acidic substances).

Classification of Pulpal diseases

- I. Pulpitis (Pulp inflammation).
- II. Necrosis.
- III. Pulp Calcifications/ Pulp Stones/ Denticles.
- IV. Internal resorption

Pulpitis (Pulp inflammation)

Pulpitis refers to the inflammation of the dental pulp within the tooth and it is the most common cause of dental pain.

Dental caries remains the primary cause of pulpitis worldwide. Other causes (less common than caries) including trauma, restorative procedures, periodontal diseases, and iatrogenic factors.

Classification of Pulpitis:

(1) According to severity

- ☐ a-Acute (symptomatic).
- b-Chronic (Asymptomatic).

(2) According to extent of pulp involvement

- a-Focal or Partial Pulpitis.
- b-Total or Generalized Pulpitis.

(3) According to presence or absence of direct communication between dental pulp and oral environment

- a- Open pulpitis.
- b- Closed pulpitis: It may occur from trauma or excessive orthodontic forces.

Here carious lesion is absent.

(4) According to recover from the insult and damage

- a- Reversible Pulpitis.
- b- Irreversible Pulpitis.

Focal Reversible Pulpitis

(Pulp Hyperaemia)

Is a reversible inflammation of the dental pulp, causing temporary sensitivity to stimuli like cold or sweet foods, which resolves quickly once the stimulus is removed. It represents the earliest stage of pulpal inflammation from an irritant like a decayed cavity, and can be healed by removing the cause, such as treating the decay.

Pulp hyperaemia is due to active dilatation of pulpal blood vessels.

Aetiology: Any of previously mentioned etiological factors can cause pulp hyperaemia.

Clinical features:

- The tooth is sensitive to thermal changes particularly cold application. It is usually caused by cold, sweet and sour stimuli.
- The pain is sharp shooting pain.
- The pain disappears when the stimulus has been removed.
- The tooth is more sensitive to electrical pulp tester.
- In focal reversible pulpitis, the pain is typically easily localized to a specific tooth because it is triggered by a stimulus and resolves quickly once the stimulus is removed.
- However, the inability to localize the tooth perfectly can still occur, as the dental pulp lacks proprioceptive fibers and nerve fibers from a single tooth may branch to innervate other teeth, making pain sometimes diffuse.

Histopathological Features:

1. Dilated blood vessels which carry more blood to the area of injury.
2. Intact odontoblastic layer.
3. Increased permeability of the capillary walls leads to fluid exudates in the pulp tissue (oedema) that press on nerve endings causing pain.
4. Mild inflammatory cell infiltrate- mainly lymphocytes and neutrophils.

5. Presence of reactionary dentine.



Treatment:

Removal of the cause if possible

Prognosis:

Good prognosis for pulp recovery with appropriate dental intervention, such as placing a filling.

Sequelae:

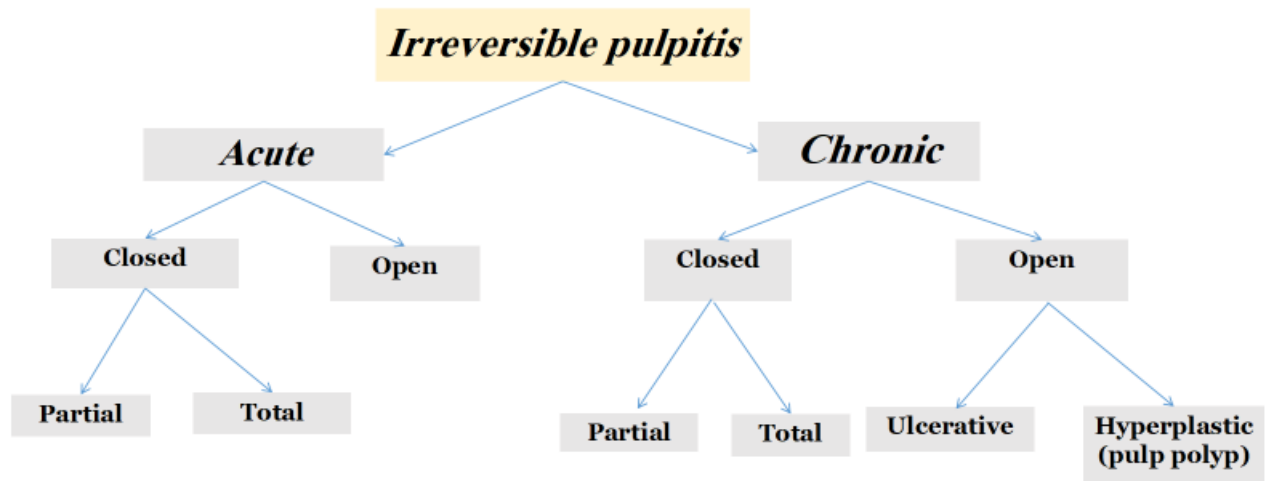
Regression (if treated before pulp damage).

Irreversible pulpitis (acute or chronic).

Irreversible pulpitis:

The process of irreversible pulpitis may be acute or chronic pulpitis.

Acute pulpitis may be a progression of focal reversible pulpitis or may present as an acute exacerbation of an already established chronic pulpitis.



Acute pulpitis

Irreversible condition characterized by acute, intense inflammatory reaction in pulp tissue.

Mode of development:

- Sequelae of focal reversible pulpitis.
- Acute exacerbation of chronic pulpitis
- As denovo condition, where inflammation is acute from beginning.

Types:

- ☐ Partial acute pulpitis confined to one horn of coronal pulp.
- ☐ Total acute pulpitis involves the whole pulp.

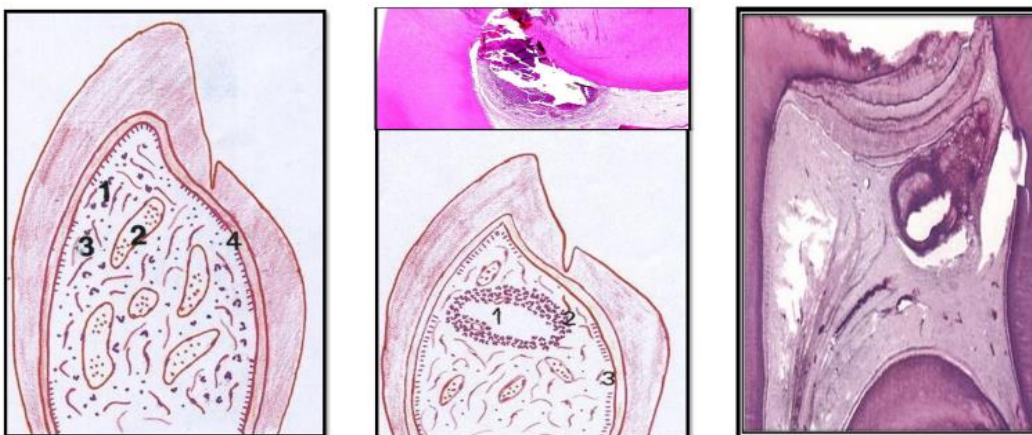
Clinical features:

- The patient with early irreversible pulpitis presented with sharp, sever pain (lancinating, throbbing) on thermal stimulation, and the pain continues after removal of the stimulus. Cold is the most uncomfortable, although heat or sweet and acidic food can cause pain.
- Later on, the pain may be spontaneous or continuous and may be exacerbated when the patient lies down.
- The tooth responds to electric pulp testing at lower levels of current.
- Mobility and sensitivity to percussion are negative.

- At early stage, the pain often can be localized easily to the individual affected tooth. With time the patient discomfort is increasing and can no more be able to identify the offending tooth.
- In the later stages of irreversible pulpitis, the Pain increases in intensity and experienced as throbbing, which keeps the patient awake at night. At this point heat increases the pain, while cold may produce relief. why?
- As the pulpal inflammation progresses, heat intensifies the response because it has an expansible effect on blood vessels. Cold tends to relive pain because of its contractile effect on vessels, reducing the intra-pulpal pressure.

Histopathological features:

1. Destruction of the odontoblastic layer.
2. Dilatation of blood vessels (Vasodilatation).
3. Edema in the connective tissue.
4. Acute inflammatory cells infiltration (polymorphonuclear leukocytes).
5. Focal areas of tissue destruction occur leading to formation of pus. Pus consists of necrotic tissues, dead and alive polymorphs and dead and alive bacteria. PMNLs in concentric layer around the pus form pulp abscess.
6. Numerous abscesses appear. Finally fuse together leading to total liquefaction of pulp forming suppurative pulpitis.



Treatment:

Root canal therapy (RCT) or tooth extraction.

Chronic Pulpitis

Chronic pulpitis is a chronic inflammation of dental pulp with little or no symptoms.

Aetiology:

1. It may follow acute pulpitis or arise de novo.
2. In case of de novo chronic pulpitis, the causative agents are usually of low virulence and the body resistance is high.
3. The general etiologic factors for pulpitis can also cause chronic pulpitis.

Types:

□ It may be closed or open.

The open may be ulcerative or hyperplastic (pulp polyp).

Clinical features:

- Pain is not a feature of chronic pulpitis due to diminished exudative inflammatory activity and decrease in intra-pulpal pressure.
- In some cases, patients may complain of intermittent dull pain.
- The reaction to thermal changes and EPT is reduced.
- The tooth is not sensitive to percussion.

Histopathological Features:

- The pulp tissue exhibits increased fibroblastic activity (to wall-off the infection).
- The pulp infiltrated with chronic inflammatory Cells (plasma cells, lymphocytes, and macrophage), fibroblasts, + new blood vessels.
- Adjacent odontoblasts are destroyed.
- Incomplete calcific barrier around the inflammatory focus.

Treatment:

Root canal therapy (RCT) or tooth extraction.

Chronic Hyperplastic Pulpitis (Pulp Polyp)

It is a hyperplasia of a chronically inflamed pulp, characterized by hyperplasia of connective tissue of the pulp in the form of polypoid mass which fungated from exposed pulp chamber in response to persistent, low-grade mechanical irritation and bacterial invasion of the pulp.

For pulp polyp to develop, 4 prerequisites should be fulfilled

- 1) A good sheltered area for protection of the newly formed tissue.
- 2) A wide apical foramen for good blood supply.
- 3) A wide pulp exposure to allow protrusion of new tissue.
- 4) Young aged patient for good proliferative power.

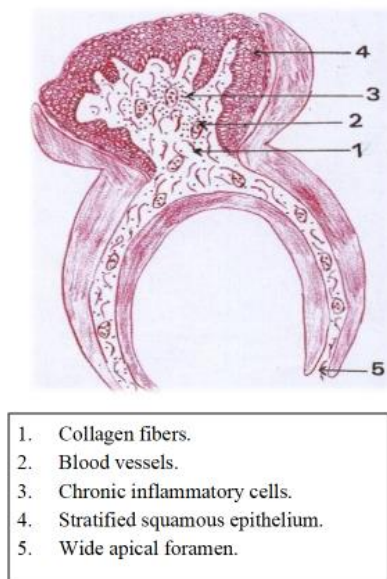
Clinical features:

- It occurs almost exclusively in children & young adults and involves teeth with large open carious cavity.
- Usually occurs in deciduous molars or the first permanent molar due to excellent blood supply and large root opening.
- There is a piece of tissue extruded from the pulp and present in a large carious lesion.
- The lesion is relatively asymptomatic (insensitive to manipulation because it contains fewer nerve endings) but may cause discomfort during mastication due to the pressure of the food bolus.
- It bleeds easily because of rich network of blood vessels.
- Should be differentiated from gingival polyp which is attached to the gingiva and is more sensitive to manipulation with less tendency to bleeding if compared to pulp polyp.

Histopathological features:

- 1) The pulp chamber is transformed into granulation tissue containing newly formed fibroblasts, collagen fibrils, blood capillaries and chronic inflammatory cells.

- 2) The granulation tissue is covered by stratified squamous epithelium derived from gingiva or desquamate epithelial cells of the oral mucosa or tongue.



Treatment:

Pulp polyp (chronic hyperplastic pulpitis) treatment depends on the tooth's condition, but generally involves surgical removal of the polyp followed by a root canal (pulpectomy) if the tooth is restorable, or extraction if too much tooth structure is lost.

Pulp Necrosis

Pulp necrosis is the death of the dental pulp typically caused by infection from a deep cavity, crack, or trauma. As this is associated with bacterial infection called **pulp gangrene**.

Etiology:

- It follows either pulpitis (acute or chronic) or traumatic injury to apical blood vessels.
- In sickle cell anemia, blockage of pulp vessels by defective RBC (cannot flow smoothly through the tiny pulp blood vessels) results pulp necrosis.

Clinical features:

- It is associated with foul odor when pulp is opened for endodontic treatment.
- Pain is absent (asymptomatic).

- Discoloration of the tooth is the first indication that the pulp is dead (due to hemolysis of red blood cells).

Histopathological features:

The pulp chamber and canals appear empty or contain structureless mass of necrotic pulp tissue. i.e. no cell, no fibers, no blood vessels.

Treatment:

RCT (root canal therapy) or tooth extraction.

Pulp Calcifications

Pulp calcifications are mineralized masses in the dental pulp and are generally classified as pulp stones or pulp canal obliteration (PCO).

Pulp stones are nodular, calcified masses appearing in either or both the coronal and root portion of the pulp. The larger calcification is called denticles. Pulp stones may form either due to some injury or natural phenomenon.

Classification of pulp stone:

1.On the basis of structure

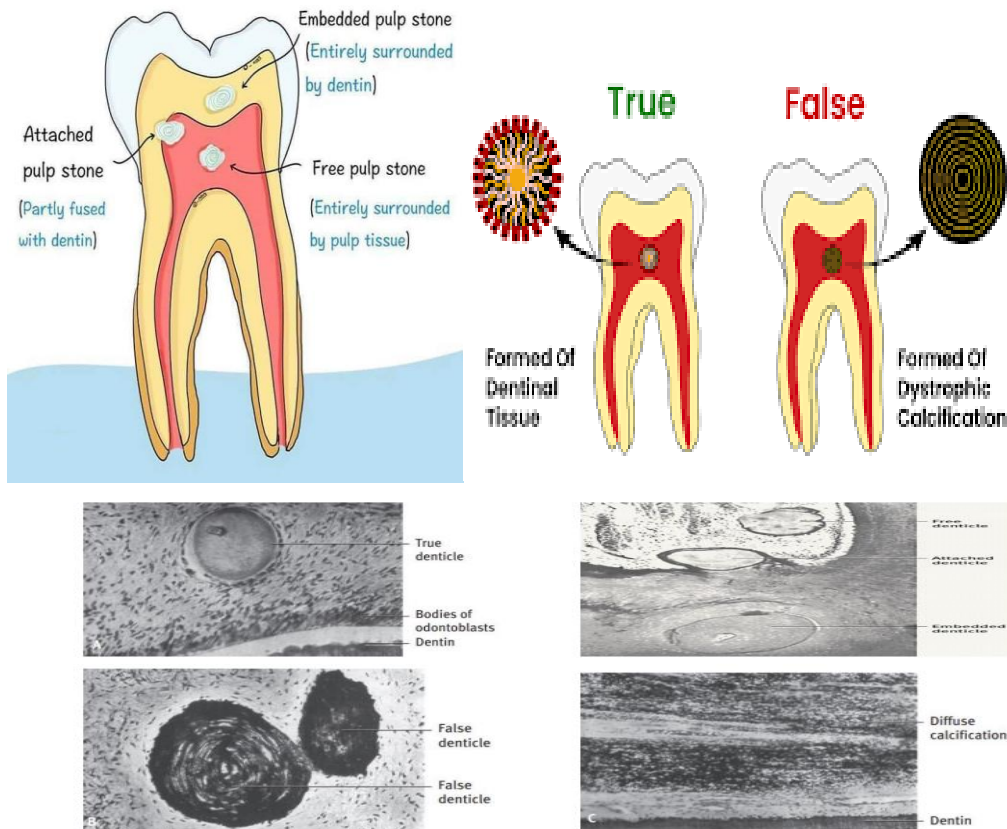
- a. True denticles:** These are rare and they are similar in structure to dentin in that they have dental tubules and surround by odontoblasts.
- b. False denticles:** These stones do not exhibit dentinal tubules but appear instead as concentric layers of calcified tissue.

2.On the basis of location

- a. Free:** These are entirely surrounded by pulp tissue.
- b. Adherent(attached):** Partly fused with dentin.
- c. Embedded:** These are entirely surrounded by dentin.

3.On the basis of the size

- a. Diffuse**
- b. Fine**



Clinical Features:

- No clinical significance. Usually, it is discovered on a radiograph only as radio opacity.
- Sometimes, it may cause pain, from mild to severe pain.
- Difficultly during root canal therapy.
- Associated with some conditions, like dentin dysplasia type II, and Ehlers-Danlos syndrome.

Treatment:

- No treatment is necessary if the tooth is asymptomatic.
- The removal of pulp calcification (pulp stones) is present another challenge in endodontic treatment, as they block the access to the root canals. The best treatment for this condition is the combination of magnification and ultrasonic tips.

True pulp stone	False pulp stone
✓ It result from undeferentated mesenchymal cell due to inductive effect of remanent of HERS.	✓ It occures at area of heamorrhage or degenerated cells
✓ It contains organic matrix with dentinal tubules and surround by odontoblasts .	✓ It consist of concentric layer of calcific material with no tubular structure
✓ Small in size	✓ Large in size

Hertwig's epithelial root sheath =HERS

Pulp obliteration **or pulp canal obliteration (PCO)**

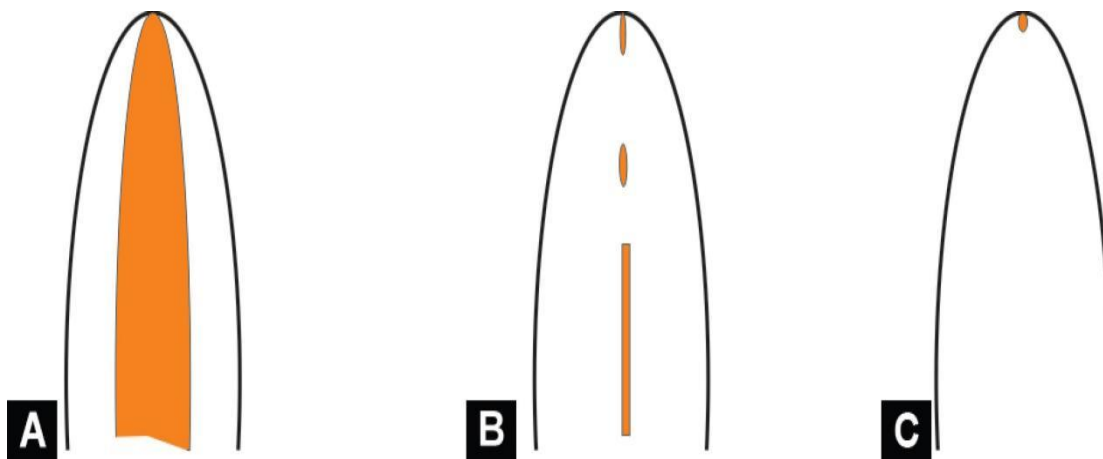
Pulp obliteration, or pulp canal obliteration (PCO), is the excessive, rapid formation of dentin within the pulp chamber and root canals, reducing or eliminating the pulp space.

Causes of pulp obliteration:

1. Trauma this is the most common cause, often following concussion or subluxation injuries, especially in young, developing teeth.
2. Aging: The process can naturally occur as part of aging.
3. Systemic conditions such as secondary hyperparathyroidism.
4. Dental anomalies such as dentinogenesis imperfecta and dentinal dysplasia.
5. Medications: Long-term use of certain medications, such as glucocorticoids (GCs).

Clinical features:

- A common sign is a yellowish or grayish discoloration of the affected tooth's crown due to increased dentin thickness.
- Routine radiographs will show a reduction in the size of the pulp chamber and narrowing or apparent complete obliteration of the root canal.
- Teeth with PCO often show lack of response to pulp vitality tests.



Progressive obliteration of pulp canal space: (A) Normal root canal; (B) Partially obliterated root canal; (C) Completely obliterated root canal

Internal resorption /pink tooth

It is an idiopathic slow or fast progressive resorptive process occurring in the dentin of the pulp chamber or root canal of teeth.

The condition is more frequently observed in male than in female subjects.

Etiology:

Unknown but is usually secondary to pulpitis.

History of trauma either a blow or restorative procedures.

Clinically:

In case of coronal resorption, the presence of a pink spot in the coronal portion of the affected tooth is due to presence of granulation tissue just under the enamel.

Radiographically:

Symmetrical, round, well defined radiolucency continuous with the pulp space. Pulp space appears enlarged; root canal outline is distorted.



Age change in the pulp:

1. Decreased pulp size gradually with age due to production of secondary dentine.
2. Decreased vascularity.
3. Decreased cellularity.
4. Increased collagen fibers.
5. Increased pulp stones.